ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/275410160

BIM Energy Modeling: Case Study of a Teaching Module for Sustainable Design and Construction Courses

Article *in* Journal of Professional Issues in Engineering Education and Practice · January 2014

DOI: 10.1061/(ASCE)EI.1943-5541.0000230

CITATIONS
12

READS
317

4 authors, including:

Rodolfo Valdes-Vasquez
Colorado State University
44 PUBLICATIONS **177** CITATIONS

SEE PROFILE

Caroline Clevenger
University of Colorado
34 PUBLICATIONS **84** CITATIONS

SEE PROFILE

Tripp Shealy
Virginia Polytechnic Institute and State University
51 PUBLICATIONS **101** CITATIONS

SEE PROFILE

Some of the authors of this publication are also working on these related projects:

Project Climate Change Literacy View project

Project Public Sociology View project

All content following this page was uploaded by Tripp Shealy on 25 May 2016.

The user has requested enhancement of the downloaded file.

BIM Energy Modeling: Case Study of a Teaching Module for Sustainable Design and Construction Courses

Anderson M. Lewis[1]; Rodolfo Valdes-Vasquez, Ph.D., A.M.ASCE[2]; Caroline Clevenger, Ph.D., M.ASCE[3]; and Tripp Shealy[4]

Downloaded from ascelibrary.org by Vpi & Su Tech Services/Serials Recvg on 05/25/16. Copyright ASCE. For personal use only; all rights reserved.

Abstract: Energy modeling (EM) facilitates analysis and comparison of energy use across design configurations. EM software can serve as a decision-making tool for professionals throughout various stages of building delivery. This case study describes methods to expose students, who are enrolled in a sustainable design and construction course, to building information modeling (BIM)–based EM capabilities and limitations. The case study focuses on documenting the implementation of a teaching module focused on energy modeling in which students modify location, insulation, and window glazing of a single family home preprogramed into BIM software. After instructor demonstration, students perform an EM analysis for additional locations as homework and answer a question set based on their EM results. The results indicate that students had minor issues interpreting the EM results because they did not fully examine the outputs produced from the energy model to inform their answers on the teaching module homework assignment. Overall, however, research findings suggest that implementing such a teaching module can improve student confidence and perception of EM use in construction delivery decisions. This paper provides an example for educators to include BIM EM or similar teaching modules into sustainable design and construction-related courses. **DOI: 10.1061/(ASCE)EI.1943-5541.0000230.** *© 2014 American Society of Civil Engineers.*

Author keywords: Sustainable development; Energy modeling; Undergraduate study; Building information modeling (BIM).

Introduction

Building information modeling (BIM) is a process that utilizes 3D computer rendering software capable of generating visual feedback for stakeholders to document building design and simulate construction and operation of a facility (Gordon et al. 2009). The architectural, engineering, and construction (AEC) industry is quickly adopting BIM as a design and project management tool because such processes have the potential to increase productivity during the project life cycle. For instance, the owner can use BIM to recognize project needs, the design team can analyze several project scopes, the general contractor can enhance coordination with vendors and suppliers, and the facilities manager can analyze the performance of the facility during the operation and decommissioning phases (Bryde et al. 2013). A report from McGraw-Hill (2012) Construction indicates BIM adoption rates in the U.S. AEC industry more than tripled between 2007 and 2012.

With BIM usage continuing to grow, a corresponding need for competent building information modelers is also needed (Azhar 2011; Ahn et al. 2013). Graduates with BIM education bring new implementation strategies to the delivery process (Gordon et al. 2009). Employers recognize the benefits of those competent in BIM software and have begun to ask for BIM skills as a requirement for future employees (Gordon et al. 2009). To insure educators meet the demand for BIM-trained professionals and a more marketable work force, training materials and technical support related to BIM software needs further development (Gu and London 2010).

Despite all the benefits to engineering and construction students, creating a curriculum that includes BIM is not straightforward. A study performed by Sabongi (2009) found that fewer than 1% of schools teach BIM as a standalone course, and only 9% address BIM in existing courses. As schools continue to increase their capacity for teaching BIM, the practice is far from comprehensive, and the rate of integration is much slower than industry needs (Sabongi 2009).

Multiple factors inhibit BIM integration into existing curricula. Student curriculum is full. Adding additional courses would require removing existing ones. Faculty time and resources are limited. Educational resources and time to develop course material are lacking (Sabongi 2009). Uncertainty regarding dominant technologies makes it difficult for institutions to commit to teaching certain BIM programs. Beyond these implementation factors, BIM is used for a wide range of analysis, including energy modeling, and is providing new opportunities for young professionals (Russell et al. 2014).

Another technology that is helping designers improve their decision making pertaining to the built environment is energy modeling. Energy modeling is normally considered to be the process of predicting a building's energy performance through software analysis, whereas an energy model is a computerized representation of a building and its properties that is used to perform energy analysis calculations [American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) 2005]. A variety of energy modeling software products exist, some of which perform specific analysis on single aspects of building

[1]Graduate Research Assistant, Dept. of Construction Management, Colorado State Univ., Fort Collins, CO 80523. E-mail: amlax3@colostate.edu

[2]Assistant Professor, Dept. of Construction Management, Colorado State Univ., Fort Collins, CO 80523 (corresponding author). E-mail: rvaldes@colostate.edu

[3]Associate Professor, Dept. of Civil Engineering, Univ. of Colorado Denver, Denver, CO 80217. E-mail: caroline.Clevenger@ucdenver.edu

[4]Graduate Research Assistant, Glenn Dept. of Civil Engineering, Clemson Univ., 208 Lowry Hall, Clemson, SC 29634. E-mail: eshealy@g.clemson.edu

Note. This manuscript was submitted on December 29, 2013; approved on October 15, 2014; published online on December 4, 2014. Discussion period open until May 4, 2015; separate discussions must be submitted for individual papers. This paper is part of the ***Journal of Professional Issues in Engineering Education & Practice***, © ASCE, ISSN 1052-3928/C5014005(6)/$25.00.

design (such as daylighting or mechanical systems), whereas other software perform whole building analysis, which looks at the overall performance of a building (Crawley et al. 2008). Energy models, when used properly, have the potential to lead to more efficient designs, lower costs, and result in better value for a building's owner by allowing the energy implications of multiple designs to be efficiently compared and examined. Knowing how to run energy simulations and being familiar with their results can be a valuable skill in the AEC industry because it allows users to more quickly analyze design options to provide better value and better performing buildings to the owner. Energy modeling itself tends to be an involved and error-prone process because it requires identifying and entering numerous inputs into the software program. However, leveraging geometries and information imbedded in BIM models can greatly hasten and simplify the energy simulation input process.

Teaching energy modeling (EM) potentially presents similar challenges to those associated with teaching BIM (Sabongi 2009). Extensive credit requirements for graduation deter new courses being added. Faculty, for instance, lack the time and educational resources to overcome the initial complexity and uncertainty regarding adopting new technologies and processes in the classroom. To overcome such integration challenges in these and related areas, the authors suggest adding stand-alone teaching modules, similar to the one developed for this research, to existing classes instead of creating distinct courses that teach specific tools, including EM or even BIM, to construction students. Providing students the opportunity to interactively use and learn BIM and EM software programs and apply concepts from their sustainable design and construction course is an engaging method to familiarize students with the capabilities and potential uses of these tools.

Furthermore, certain EM tools are BIM-based or BIM compatible. These products allow users to extract information directly from a BIM model and import this information into an EM. BIM-based EM can save users considerable time by minimizing the amount of data they have to manually enter into an EM. This can allow for more EMs to be run in a shorter amount of time, which, in turn, gives the user the opportunity to explore the cost and energy implication of more design options. Lastly, BIM-based energy models allow the user to see a 3D visual model to make sure that the owner's aesthetic and functionality requirements are still met.

The purpose of this paper is to discuss implementation of a teaching module, which introduces students to BIM-based EM capabilities. Students learn the process of running energy simulations and how energy modeling results can better inform their decisions related to energy efficiency design. An introductory BIM EM teaching module was created that can serve as a model or is replicable in other engineering or construction courses. The overall objective, beyond gaining familiarity with the energy modeling process, is to provide students an opportunity to apply and synthesize concepts related to specific sustainable design and construction options, such as the impact of insulation and window type in varying climate zones. In addition, this teaching module demonstrates that EM is a decision-making tool for analyzing the energy impacts across design options. Before introducing the implementation of the teaching module, the next section provides a brief overview of the teaching module development.

Teaching Energy Modeling

Teaching the relationship between building science and energy use is complex. Every decision involves tradeoffs. Using energy modeling to explore the interactive effects of design and construction decisions across building types and climates can help students become more aware of these tradeoffs and more adept in decision making. In fact, lack of continuous energy analysis during the design process can decrease building energy performance, and retroactive modifications to meet higher performance are costly (Azhar et al. 2011). BIM can rapidly produce energy outputs that enable design teams to analyze and compare the most cost-effective, energy-efficient options (Schade et al. 2011). As BIM technology and modelers' skills continue to improve, project design teams will better integrate EM simulations into the delivery process (Froese 2010). BIM software now includes life-cycle cost and life-cycle assessment technology, allowing project teams to be better informed during the decision-making process (Azhar 2011).

A case study by Kim and Shim (2011) describes one student's experience to integrate green building design concepts into a semester long *Revit* course project. One of the major challenges faced by this student was a lack of prior knowledge of green building design measures in addition to a lack of familiarity with the different *Revit* functions. The student first produced a baseline model then created design alternatives. Using the *Green Building Studio (GBS)* energy simulation tool, the student tested each design alternative measuring the cost impact on whole building performance. Through these iterations (and based on the energy simulation results), the student was able to predict annual savings by adjusting building orientation and window shading, changing wall and roof materials, and using low-flow flush toilets. Additional savings were predicted through modeling new high-efficiency lighting and HVAC systems.

In this case study, the student learned the uses and advantages of using BIM to integrate green building measures into building designs while progressing through the semester. Unlike the aforementioned case study, the present teaching module was developed for students with limited or no experience with BIM products but who were familiar with sustainable building concepts such as internal loads, insulation properties, and integrated design.

Development of the Teaching Module

For this research, a BIM EM teaching module was developed for the Sustainable Design and Construction (CON 476) course at Colorado State University (CSU). The CON 476 course is an elective for upper level undergraduate and graduate students. During fall semester of 2013, 20 students enrolled, including civil engineering, construction management, and interior design majors. The course explores how to integrate environmental, social, and economic tradeoffs into project delivery. This course also includes an introduction to the major components of building energy usage, healthy buildings, cultural and social aspects, and natural resource consumption. In a similar EM teaching module performed in previous years, *eQUEST* had been the energy modeling software used to familiarize students with the most basics concepts about how to create an energy model and how the results can be used as a decision-making tool (Knox and Clevenger 2014). This teaching module was designed to take an entire 3-h class period.

Software

Building on this previous work, the teaching module developed for this research uses *Revit* in conjunction with *GBS*. *Revit* was chosen for the platform for this assignment because it is the most widely used BIM software in the United States. In addition, free temporary licenses are available to students. According to Autodesk (2014), GBS is "a flexible cloud-based service that allows you to run building performance simulations to optimize energy efficiency and to work toward carbon neutrality earlier in the design process." However, registering for GBS and uploading previously developed

J. Prof. Issues Eng. Educ. Pract., 2015, 141(2): C5014005

Downloaded from ascelibrary.org by Vpi & Su Tech Services/Serials Recvg on 05/25/16. Copyright ASCE. For personal use only; all rights reserved.

files into the cloud-based software was a challenging step for students.

Within *Revit*, two types of energy simulations are possible. This module uses the building element–based energy simulation, which analyzes the physical properties of individual building components. This simulation provides more precise results, but requires more data inputs and can be more challenging to learn than the conceptual mass alternative. The more basic conceptual mass–based simulations only account for geometry, size, and location. This simulation is better suited for early design phases.

Model

Because of time restrictions and varying levels of experience among students using *Revit* and to maintain consistency among student results, the instructor created a model that was distributed to students. In this instance, a residential model was created because of its simplicity. More complex models can be used for similar teaching modules; however, more complex models are likely to take more time to edit and run energy simulations. In addition, previous research has indicated that interoperability issues and lack of BIM standards may result in errors and accuracy issues (Raftery et al. 2011; Bryde et al. 2013). Although BIM data exchange standards such as Industry Foundation Classes Extensible Markup Language (ifcXML) and Green Building Extensible Markup Language (gbXML) schema do exist, no information infrastructure has been unanimously adopted across the entire AEC industry (Hitchcock and Wong 2011). Although this residential model was relatively simple, in theory, the act of importing a more complex model might present greater difficulty. In courses in which students are more advanced with using the BIM software at hand, students may be asked to create their own models; however, if students create their own models, the question sets will have to be more open-ended.

Implementation

The purpose of this teaching module is to (1) expose students to energy modeling tools and practices; (2) demonstrate energy modeling input requirements to provide context for processes and results; and (3) provide opportunities to virtually compare and contrast potential energy impacts of design and construction changes. The overall implementation of this teaching module is broken down into four phases: preassessment, in-class exercise, homework, and postassessment survey.

Phase 1: Preassessment

Phase 1 consisted of three parts that occurred one week before class. First, a survey was sent out to measure students' perceptions about sustainability and the potential for BIM to facilitate the achievement of energy goals. The preassessment survey also measured student familiarity with BIM and EM programs. Second, one week before the teaching module, students were given an outline of the EM assignment. Last, students were asked to sign up for an Autodesk 360 account and a *GBS* account. The instructor created a tutorial to guide students through this process.

Phase 2: In-Class Activity

Before introducing the BIM energy model exercise, an Autodesk representative was invited as a guest speaker. The representative, a program manager for the education division, discussed the evolution of BIM and how predictive analysis tools, in particular EM tools, can be used to guide the decision-making process. The guest speaker also briefly discussed the Autodesk product line capabilities and how to best use each performance analysis.

After the guest speaker presentation, the instructor gave students a brief overview of the assignment, followed by an EM demonstration. Students were asked to open a previously created *Revit* file that was a model 1800 SF ranch-inspired house. Students were expected to run energy simulations for three different locations: Fort Collins, Colorado; Fairbanks, Alaska; and Miami, Florida. These locations were selected to contrast energy-saving strategies in different climate zones. For each location, an EM analysis was performed using only the baseline model.

Alternative 1 increases the wall insulation from R-8, from the baseline analysis, to R-16. Alternative 2 upgrades windows from a 1/8-in. single pane window to low emissivity, double-pane 1/4-in. windows. Each EM analysis was distinct and did not build on previous improvements. Enhanced insulation and windows were chosen for EM runs because they are illustrative of passive design strategies that impact building performance and are relatively easy to change through EM's program inventory.

Students were guided through building the baseline model. This included entering predetermined assumptions about building location, building type, lighting assumptions, HVAC systems, building envelope material, occupancy schedule, and ground plane. In class, the students were guided through the first EM set for Fort Collins. Students were then asked to run an energy simulation for two different locations, Fairbanks and Miami, as homework. The students were given a detailed set of instructions to complete the remaining energy simulations. Fig. 1 is an example of the instructions followed by students.

Phase 3: Homework

Students answered a question set comparing their EM results (see Appendix S1) to sustainability concepts learned during the semester (such as passive solar design strategies). Although students were expected to run all the simulations on their own, owing to the complexity of this assignment and technical difficulties with Autodesk 360 account sign-up and laboratory computers, it was hard to ensure all students would generate the same results. The instructor's EM results were given out as a guide for answering homework questions to ensure that all students had the same results to answer homework questions.

Phase 4: Postassessment Survey

A follow-up survey measured students' perceptions of BIM EM as a valuable decision-making tool and student confidence in using *Revit* EM functions for future projects. Questions asked student confidence in comprehending EM results and if these results could guide decision makers between building materials and energy efficiency tradeoffs.

Results and Analysis

Preassessment Survey

The preassessment survey was administered a week before the class exercise to measure students' perceptions regarding sustainability and the importance of the role BIM can play in achieving sustainability and their familiarity with BIM programs and EM functions. A total of 16 students responded to the survey. As expected, from students enrolled in a sustainable design and construction course, students almost unanimously agreed with the statement "energy conservation in buildings is important for future generations." In

J. Prof. Issues Eng. Educ. Pract., 2015, 141(2): C5014005

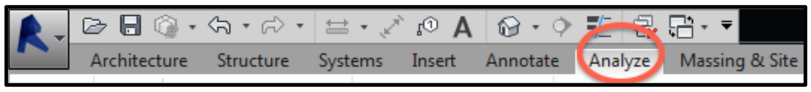
Downloaded from ascelibrary.org by Vpi & Su Tech Services/Serials Recvg on 05/25/16. Copyright ASCE. For personal use only; all rights reserved.

Downloaded from ascelibrary.org by Vpi & Su Tech Services/Serials Recvg on 05/25/16. Copyright ASCE. For personal use only; all rights reserved.

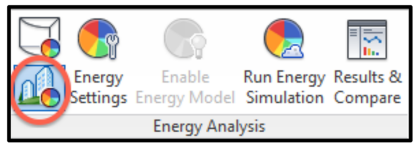
INSTRUCTIONS

Creating Input Model -

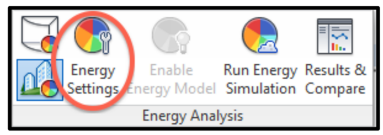
1. Click on the "Analyze" tab at the top of the screen.



2. Under the Energy Analysis section (in the upper-right corner of your screen) make sure that "Building Element Mode" is highlighted (instead of Conceptual Mass Mode). (Illustrated below)



3. Next, Click on "Energy Settings" (to the right of building element mode)



- The window below will appear

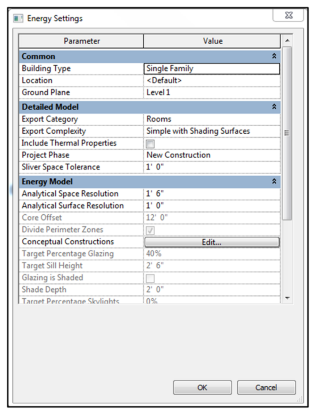


Fig. 1. Example of instructions (images courtesy of Autodesk, with permission from Autodesk)

total, 15 of 16 students agreed or strongly agreed with the statement "Building information modeling (BIM) energy modeling is an important tool for designing energy efficient buildings." Students also believed BIM energy modeling tools help facilitate communication among stakeholders (i.e., architect, engineer, contractor, owner) on sustainable building projects. Table 1 illustrates students' perceptions of BIM EM function usefulness as a decision making tool for various design options. The survey implemented a 7-point Likert-scale.

Table 1. Mean, Median, and Mode for Responses to Presurvey Question

BIM energy modeling can be used as a decision-making tool pertaining to building	Mean	Median	Mode
Orientation	6.19	6	7
Geometry	5.94	6	6
Envelope properties	6.00	6	6
Lighting	6.19	6	7
Mechanical systems	6.13	6	6

The survey results indicated that seven of the students (43.8%) had previous experience with BIM software programs: Autodesk's *Revit*, AutoCAD, Navisworks, Solid Works, *Vico Office*, Autodesk *Structural Analysis Professional*, *3D Max*, and *Google Sketch-Up*. Only two students (12.5%) indicated they used the Energy Modeling function of *Revit* in the past year. Despite the lack of *Revit* EM experience, students indicated varying levels of confidence in their ability to use the *Revit* EM function.

Homework

Responses from the homework assignment suggest students can comprehend and interpret EM results. Students identified differences in building performance across distinct location and design iterations. For example, students realized that gas usage went up significantly in Fairbanks over the other two locations because the model used gas heating. Fairbanks is on average quite a bit colder than the other two locations. Students sporadically missed a variety of questions. However, it is hard to determine if this was attributable to a lack of comprehension of the energy modeling

J. Prof. Issues Eng. Educ. Pract., 2015, 141(2): C5014005

results or other factors. For example, the question that was most often missed was "When comparing the baseline EM results for Fairbanks, AK and Fort Collins, CO, why might 'Life Cycle Electricity Use' be relatively higher in all runs for the building located in Fairbanks, AK?" Many students said that it was attributable to the differences in lighting loads because Fairbanks receives fewer hours of sunlight in the winter. Although this is a valid argument, the EM results did not reflect this hypothesis because the lighting schedules were, in fact, the same for both facilities regardless of hours of daylight. The EM results made it clear that the largest discrepancy in energy usage between Fort Collins and Fairbanks was attributable to the HVAC heating.

Another commonly missed question was, "Why might the 'Life Cycle Fuel Use' be higher for the Window EM run for Miami than the Baseline EM run?" This question requires students to apply sustainable concepts learned in the course. The majority of the wrong answers were not informed by the EM results. One of the incorrect answers attributed the Window EM run as contributing to the extra fuel use because they were "less insulated," when they were in fact more insulated. The answer sought for the question is that the lower solar heat gain coefficient for the windows in the Window EM run reduced the solar gains of the low-angled winter sun, causing the mechanical systems to have to heat more than they otherwise would have.

Postassessment Survey and Comparison

After students participated in the BIM EM teaching module and completed the homework, a postassessment survey collected student perception and confidence levels in using BIM EM. Fourteen responses (70% response rate) were collected. Similarly to the first preassessment survey, respondents were found to have high levels of agreement with the statement "Building Information Modeling (BIM) energy modeling is an important tool for designing energy efficient buildings," with 57% agreeing with this statement and 43% strongly agreeing with this statement. In this instance, the results indicated that the teaching module had no impact on students' perception of the importance of BIM EM functions for designing energy efficient buildings.

When comparing responses from the preassessment to postassessment survey, respondents indicated a noticeable increase in confidence in their abilities to use *Revit* Energy Modeling functions. The mean scores for this item increased from 3.00 to 4.71 out of 7. Although there were two fewer respondents in the second survey, if these respondents had taken the survey and responded "completely unconfident" (0 points), the score for this item would increase 0.75 points. Students' confidence levels increased because of their exposure to *Revit*'s EM functions during the teaching module and homework exercise. Fig. 2 summarizes the change in student confidence levels in using BIM EM functions from the preassessment survey to the postassessment survey.

When asked to what degree respondents agreed with the statement "I can use BIM EM results to guide my design decision-making process and defend my decision," the average rating was 5.57 out of 7. Only one respondent did not agree to some level with the previous item. When asked to indicate to what level they agreed with the statement "this homework helped familiarize me with how to change/update the following inputs required for an energy model" for location, insulation, and window glazing, there were high response scores, with mean scores of 5.86, 5.64, and 5.50, out of 7, respectively. These results are illustrated in Fig. 3.

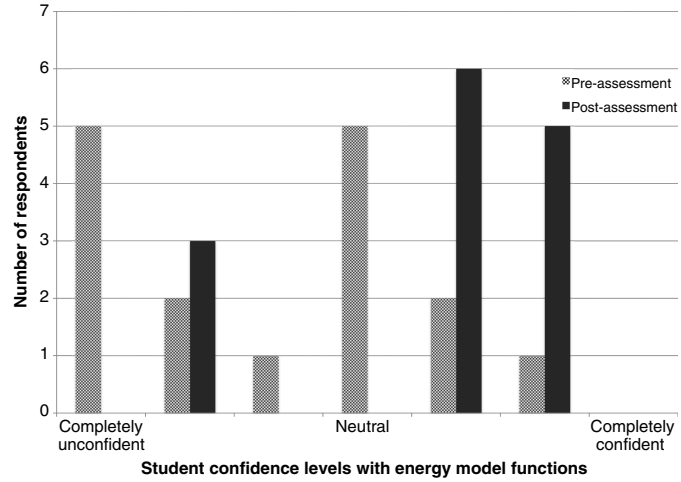


Fig. 2. Confidence levels with *Revit* EM functions: preassessment versus postassessment surveys

Lessons Learned

This teaching module provided the authors with multiple valuable lessons. First, results suggest that the developed teaching module is an effective way to introduce students to using BIM-based energy modeling and may serve as a useful model for developing teaching material in related areas. The teaching module as presented allows students the ability to see how energy simulation tools can leverage BIM models. These models produce information that guide decisions and provide students a foundation to continue exploring BIM-based sustainability analysis on their own.

The second major lesson learned is that orchestrating a teaching module that relies on software modeling with numerous and variable inputs can be logistically challenging. The in-class portion of this teaching module required the instructor to resolve computer and software issues as they arose for each student. This situation greatly slowed down the progress of the class. Instead of each student following along step by step, a more effective method may be to show the students the steps to running a BIM-based energy simulation, then allowing them to follow step-by-step instructions on their own to complete the assignment.

Finally, results of the surveys show that prior to the activity, students already perceived EM as important to the overall sustainable

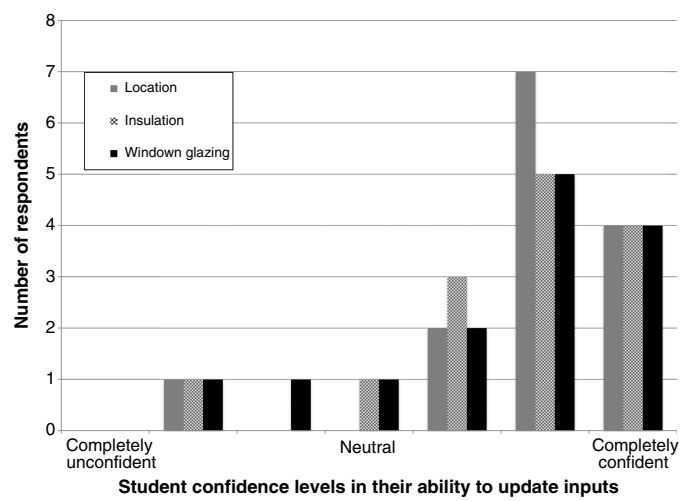


Fig. 3. Familiarity with updating inputs for an energy model

J. Prof. Issues Eng. Educ. Pract., 2015, 141(2): C5014005

Downloaded from ascelibrary.org by Vpi & Su Tech Services/Serials Recvg on 05/25/16. Copyright ASCE. For personal use only; all rights reserved.

decision-making process. Such results are encouraging because students recognize EM as a significant contributor to the overall decision-making process of sustainable building design. However, exposing students to an EM teaching module further develops students' perception and willingness to use EM on future projects. In addition, students view EM as a valuable communication tool between project stakeholders. Results from the postsurvey indicate that students' confidence in EM navigation and decision making between material choices improved after completion of the exercise.

Summary and Conclusions

This paper provides a discussion about the development and implementation of a *Revit*-based Energy Modeling (EM) teaching module, where the aim was to familiarize engineering and construction students with the process, capabilities, and limitations of energy modeling to improve decision making. Students who understand BIM EM and the decision process can market these skills to future employers. Students learned building element–based energy simulations using a prebuilt model. Students made changes to wall type, insulation levels, and window glazing. The overall objective, beyond gaining familiarity with the energy modeling process, was to provide students an opportunity to apply and synthesize concepts related to one aspect of sustainable design and construction through relatively quick comparison of cost and energy impacts of various material features. Students' perception about energy modeling can play an important role in sustainable design decisions. After the lecture and homework, students report a higher level of confidence with the EM process.

In addition, this paper provides a template for implementing similar introductory EM or similar teaching modules into sustainable engineering and construction-related courses. The findings suggest an EM teaching module may benefit students' decision-making skills and increase their confidence in construction modeling technology. This teaching module provides educators with a framework for introducing the next generation of designers and builders to key energy modeling concepts. Some of the most promising opportunities for further research include implementing and comparing this teaching module in other sustainable design and construction-related courses. Additional future research could investigate the different ways in which students experience, or think about, the developed energy modeling assignment.

Acknowledgments

The authors acknowledge the support Michael Bergin from Autodesk Education during the development and implementation of this teaching module.

Supplemental Data

Appendix S1 is available online in the ASCE Library (www.ascelibrary.org).

References

Ahn, Y., Lee, N., and Cho, C. (2013). "Building information modeling: Systematic course development for undergraduate construction students." *J. Prof. Issues Eng. Educ. Pract.*, 10.1061/(ASCE)EI.1943-5541.0000164, 290–300.

ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers). (2005). "Chapter 32: Energy estimating and modeling methods." *ASHRAE handbook—Fundamentals*, Atlanta.

Autodesk. (2014). *Green building studio*, San Francisco, ⟨https://gbs.autodesk.com/GBS/⟩ (Dec. 8, 2013).

Azhar, S. (2011). "Building information modeling (BIM): Trends, benefits, risks, and challenges for the AEC industry." *Leadership Manage. Eng.*, 10.1061/(ASCE)LM.1943-5630.0000127, 241–252.

Azhar, S., Carlton, W. A., Olsen, D., and Ahmad, I. (2011). "Building information modeling for sustainable design and LEED® rating analysis." *Autom. Constr.*, 20(2), 217–224.

Bryde, D., Broquetas, M., and Volm, J. M. (2013). "The project benefits of building information modeling (BIM)." *Int. J. Project Manage.*, 31(7), 971–980.

Crawley, D. B., Hand, J. W., Kummert, M., and Griffith, B. T. (2008). "Contrasting the capabilities of building energy performance simulation programs." *Build. Environ.*, 43(4), 661–673.

Froese, T. M. (2010). "The impact of emerging information technology on project management for construction." *Autom. Constr.*, 19(5), 531–538.

Gordon, C., Azambuja, M., and Werner, A. M. (2009). "BIM across the construction curriculum." *2009 ASC Region III Conf.*, Associated Schools of Construction, Windsor, CO.

Green Building Studio (GBS) [Computer software]. San Francisco, Autodesk.

Gu, N., and London, K. (2010). "Understanding and facilitating BIM adoption in the AEC industry." *Autom. Constr.*, 19(8), 988–999.

Hitchcock, R. J., and Wong, J. (2011). "Transforming IFC architectural view BIMs for energy simulation." *Proc., 12th Building Simulation Conf.*, International Building Performance Simulation Association, Ottawa, ON, Canada.

Kim, S., and Shim, E. (2011). "A case study in application of green building strategies using building information modeling into a construction education." *2009 ASC Region III Conf.*, Associated Schools of Construction, Windsor, CO.

Knox, M., and Clevenger, C. (2014). *Energy modeling: A tutorial and introduction to eQUEST*, Center for Sustainable Engineering Electronic Library, Syracuse Univ., Syracuse, NY.

McGraw-Hill Construction. (2012). *The business value of BIM in North America: Multi-year trend analysis and user ratings (2007-2012): Smart market report*, McGraw-Hill, New York.

Raftery, P., Keane, M., and O'Donnell, J. (2011). "Calibrating whole building energy models: An evidence-based methodology." *Energy Build.*, 43(9), 2356–2364.

Revit [Computer software]. San Francisco, Autodesk.

Russell, D., Cho, Y., and Cylwik, E. (2014). "Learning opportunities and career implications of experience with BIM/VDC." *Pract. Period. Struct. Des. Constr.*, 10.1061/(ASCE)SC.1943-5576.0000191, 111–121.

Sabongi, F. J. (2009). "The Integration of BIM in the undergraduate curriculum: An analysis of undergraduate courses." *Proc., 45th Annual Conf. of ASC*, Associated Schools of Construction, Windsor, CO.

Schade, J., Olofsson, T., and Schreyer, M. (2011). "Decision-making in a model-based design process." *Constr. Manage. Econ.*, 29(4), 371–382.

Downloaded from ascelibrary.org by Vpi & Su Tech Services/Serials Recvg on 05/25/16. Copyright ASCE. For personal use only; all rights reserved.

View publication stats